

CS 4300 Final Project - PEAS Assessment
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Introduction

The environment I've chosen to create is based around the Viking board game, "Hnefetafl". The game has defenders and attackers. On the defenders team, one piece is designated as the 'king' whose objective is to escape to any edge of the board to win, and the defenders are used to protect it as it accomplishes that goal. The attackers are trying to trap the king by surrounding it on all 4 sides. Both teams can surround an enemy on two opposite sides in order to capture it and remove it from the board.

PEAS Assessment

Performance Measure

The performance measure is simply based on whether the defender wins or loses.

- Win: 1
- Lose : 0

Environment

Observability: Fully Observable

- The entire board and the positions of all pieces are visible to the agent

Uncertainty: Deterministic

- Each action has a predictable outcome based on the rules of the game

Duration: Sequential

- The agent must make a series of moves and the sequence of actions impacts the final outcome

Stability: Static

- The environment doesn't change unless the agent or opponent makes a move

Granularity: Discrete

- The game operates on a grid-based board, with discrete turns and movement options

Participants: Multi-agent

- The environment involves two players (defender and attacker)

Knowledge: Known

- The rules of Hnefatafl are fully known to the agent, there are no unknown variables

Actions

- Move
 - Move any piece to a valid adjacent tile according to the game rules. The king piece has additional movement rules.

Percepts

- Board State
 - The current positions of all pieces
- Available Moves
 - The agent perceives which moves are legal based on the game's rules
- Capture Status
 - The agent perceives which pieces are captured after having made a move